

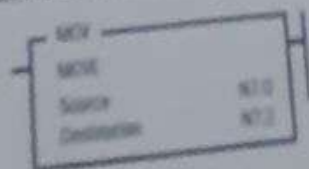


7	LD 10.1 LDN 10.3 A 10.4 LDN 10.2 AN 10.5 OLD ALD LD 10.7 O 10.6 ALD = Q0.0	
8	LD Q0.0 SIN VD16, VD20	
9	LD 10.0 ON 10.1 LDN 10.3 A 10.4 LDN 10.2 AN 10.5 OLD ALD LD 10.7 O 10.6 ALD = Q0.0	
10	LD Q0.0 SQRT AC0, AC1	

Q2 (10 Marks)

Select the best correct answer, and write it in the table below.

1- In the following math function, the N7:2 is the register that:



a- Move the data
c- Complete the data

b- Delete the data
d- Stored the data



2- In the following ladder diagram the output bit 0.013, will be activated if



- a- The value in N7:5 is less than the value in N7:10
 - b- The value in N7:5 is greater than the value in N7:10
 - c- The value in N7:5 is equal to the value in N7:10
 - d- The value in N7:10 is less than the value in N7:5
- 3- The instruction used in PLCs which save program scan time by omitting a program segment until needed is:

- a- The MOVE instruction.
- b- The ADD instruction
- c- The OR instruction
- d- The JUMP instruction

In the following ladder diagram, if input word 12, bit 10 is set then:



- a- The value in N7:3 is added to the value in N7:4 and the result is stored in N7:20
- b- The value in N7:3 is added to the value in N7:4 and the result is stored in N7:3
- c- The value in N7:20 is added to the value in N7:4 and the result is stored in N7:20
- d- The value in N7:20 is added to the value in N7:4 and the result is stored in N7:3

4- An example of discrete (digital) control is:

- a- Turning a lamp ON or OFF
- b- Varying the amount of a pressure system
- c- Varying the brightness of a lamp
- d- Controlling the speed of a fan

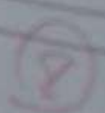
5- In the following ladder diagram, if input word 12, bit 10 is set then:



- a- The value in N7:3 is divided by the value in N7:4 and the result is stored in N7:3.
- b- The value in N7:3 is divided by the value in N7:4 and the result is stored in N7:20.
- c- The value in N7:4 is divided by the value in N7:3 and the result is stored in N7:20.
- d- The value in N7:20 is divided by the value in N7:4 and the result is stored in N7:3.

Q. NO	1	2	3	4	5
Ans	D	C	D	A	C

Good Luck



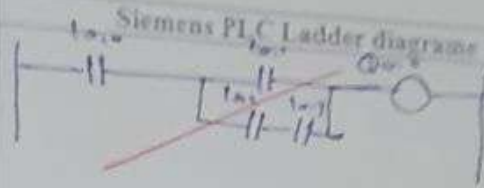


Q1 (20 Marks)

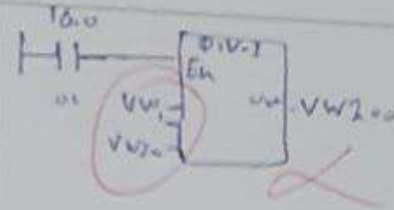
Draw the Ladder diagrams for the following instruction list shown in the table using a Siemens S7-200 PLC.

No. Siemens PLC Instruction list

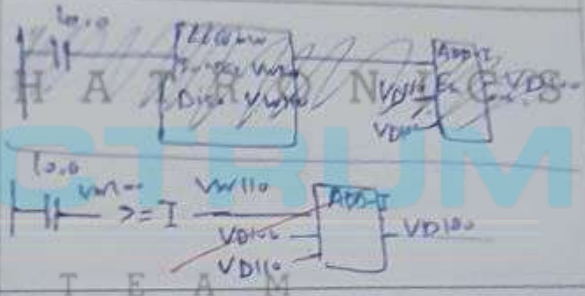
1
LD I0.0
LD I0.1
LD I0.2
A I0.3
OLD
ALD
= Q0.0



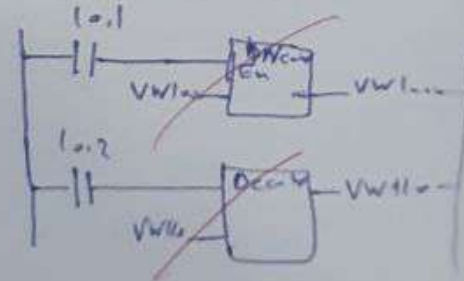
2
LD I0.0
DIV-1 VW10, VW200



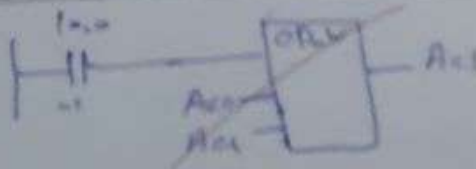
3
LD I0.0
AI >= VW100, VW110
ADD-1 VD100, VD100



4
LD I0.1
INCW VW100
LD I0.2
DECW VW110



5
LD I0.0
ORW AC0, AC1



6
LD I0.0
= Q0.0
S Q0.1
LD I0.1
R Q0.1

